

Project TDT-Stadium: A Full-Stack E-Commerce Platform for Football Equipment

Student Name: Tran Dinh Thong

Student ID: 001340570

Supervisor: Tran Trong Minh

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1. Introduction

1.1. Project Topic

This project involves the design and development of an innovative, full-stack intelligent e-commerce web platform selling football equipment. The system is based on the MERN stack (MongoDB, Express.js, React.js, Node.js), which combines conventional e-commerce features such as secure payment processing with PayPal, user authentication, and product inventory management with cutting-edge Artificial Intelligence (AI). In particular, the system uses vector embeddings and AI chatbots to offer an interactive and domain-specific virtual shopper. In today's computational environment, the incorporation of AI-powered conversational interfaces in e-commerce is quickly transitioning from a luxury to a necessity, and it can greatly improve customer engagement and provide automated support (Gentsch 2019).

1.2. Motivation

This project's driving motivation is twofold: a desire to master complex full-stack web development skills and explore the use of machine-learning-based APIs in practical applications. From an educational perspective, building a full-stack MERN application offers an excellent opportunity to gain proficiency in state management (Redux), RESTful API design, and database design. More personally, as a football player, I have found that amateur players struggle to comprehend the technical aspects of football equipment (e.g., types of boot studs for different playing surfaces, types of latex in football gloves). Creating a technology-driven application to bridge the knowledge gap is an interesting and useful challenge.

1.3. Problem Statement Summary

In contrast to the rapid expansion of global e-commerce (Statista 2023), many specialists in sporting goods stores tend to adopt static, one-size-fits-all website designs. Potential buyers of highly specialised products, like football boots and goalkeeper gloves, often find themselves without immediate expert advice. The lack of product-specific support and advice creates customer dissatisfaction, increased return rates (due to unsuitable product choices), and cart abandonment.

1.4. Originality and Significance

There are plenty of e-commerce templates (e.g. Shopify, WordPress), but few that have native integration of Retrieval-Augmented Generation (RAG) AI that is specific to a given set of products. The novelty of this project is the development of a custom e-commerce platform with an inventory of products connected to AI embeddings. This enables the chatbot to serve not only as an FAQ chatbot, but as a "virtual customer service agent" that can deliver highly targeted product recommendations, in real-time, based on current stock. This project adds to the body of research in this field by showing how small online retailers can cost-effectively use AI to simulate the knowledge of a human expert and, in turn, update the experience of a specialist retailer (Hoyer et al. 2020).

2. Problem Statement

2.1. Problem Definition

The central issue this project seeks to solve is the absence of accessible, domain-specific, and consumer-customised technical advice for navigating domain-specific sporting equipment in the online marketplace. Existing e-commerce websites primarily serve as online catalogues. There is no advice or guidance when consumers purchase highly technical football equipment, such as boots with different stud arrangements for different turf (Firm Ground vs. Artificial Grass) or goalkeepers' gloves with different latex contents.

2.2. Problem Significance

The lack of consultative support is a major issue for both consumer and retailer. For the consumer, the purchase of the wrong technical equipment can result in a poor user experience, discontent, and even the risk of injury (Smith and Jones 2022). For the retailer, this uncertainty is a major factor in the high rates of cart abandonment (which average almost 70% in e-commerce (Baymard Institute 2023) and product returns. In the footwear and sportswear industries, return rates frequently go beyond 20% and have a significant impact on retailer margins and the environment through reverse logistics (Saleh 2021). If this issue is not addressed, niche online retailers will still lose customers and sales, as they struggle to provide the type of consultative service that is typically offered in physical sports shops.

2.3. Context and Assumptions

This project is set in a contemporary online environment with amateur football players, coaches, and parents of junior football players as the target users, who seek the convenience of online shopping. The solution (an AI-powered MERN stack e-commerce system) is built with the following assumptions and constraints:

User Assumption: I assume that the customers have basic computer literacy, a stable internet connection, and will be open to chat with a conversational AI for purchase advice (Bawack et al. 2022).

Data Assumption: It is assumed that the retailer will input accurate product descriptions for products into the database. The AI vector embeddings will be as good as the text data source.

Constraints: There is a constraint on the project due to the use of third-party Natural Language Processing (NLP) APIs. The AI assistant will be limited to the store's product domain (footwear) to avoid hallucinated and irrelevant feedback.

3. Project Aim and Objectives

3.1. Project Aim

To design and develop an intelligent, full-stack e-commerce web application that provides specialized product browsing, secure purchasing, and personalized conversational guidance for amateur football players and enthusiasts, thereby enhancing the online shopping experience and facilitating accurate technical gear selection.

3.2. Project Objectives

To achieve this aim, the project will be broken down into the following measurable and time-bound objectives:

#	Objective	Timeline
1	To investigate existing specialized sporting goods e-commerce platforms and gather system requirements from potential users (minimum 5 amateur players) to identify essential features and current domain-specific challenges	(Weeks 1-2)
2	To design the overall system architecture, relational database schema, and user interface mock-ups to accommodate both standard e-commerce workflows and AI chatbot integrations	(Weeks 3-4)
3	To implement the core e-commerce functionalities, including a searchable product catalogue, shopping cart management, user authentication, and secure payment gateway integration	(Weeks 5-8)
4	To integrate a domain-specific artificial intelligence assistant capable of processing user queries and providing tailored product recommendations based strictly on the store's inventory data	(Weeks 9-10)
5	To evaluate the completed system through functional testing and user acceptance testing with at least 10 target participants, measuring system latency, usability (via a System Usability Scale questionnaire), and the contextual accuracy of the AI recommendations	(Weeks 11-12)

4. Background and Literature Review

4.1 Theoretical Underpinnings: Modern Web Architecture in E-commerce

The evolution of e-commerce platforms has increasingly shifted from monolithic architectures (such as early iterations of Magento or WordPress/WooCommerce) to Single-Page Applications (SPAs) and microservices (Biørn-Hansen et al. 2017). The MERN stack (MongoDB, Express.js, React.js, Node.js) represents a dominant theoretical framework in this shift, offering asynchronous, non-blocking I/O through Node.js and a reactive, component-based user interface via React.js. While traditional Content Management Systems (CMS) are strong in rapid deployment, they often suffer from rigid relational schemas and slower page load times as product databases scale (Kaur and Kaur 2022). MongoDB's NoSQL document structure provides the necessary flexibility to handle highly variable product attributes (e.g., varying boot sizes, stud types, and glove latex cuts) without the constraints of rigid table joins.

4.2 Review of Existing Solutions: Conversational AI in Retail

The integration of conversational AI in retail has transitioned from simplistic, rule-based chatbots to Large Language Models (LLMs). Early implementations of e-commerce chatbots relied on decision trees (e.g., IBM Watson Assistant or early Dialogflow), which are strong in handling generic FAQs (order tracking, shipping policies) but severely limited in providing dynamic, context-aware product recommendations (Gentsch 2019). Recently, Retrieval-Augmented Generation (RAG) has emerged as a solution. RAG architectures combine the reasoning capabilities of LLMs with a proprietary database of vector embeddings, allowing the AI to "read" the store's current inventory before answering (Lewis et al. 2020).

4.3 Critical Analysis of the Domain (Specialized Sports Retail)

Despite the advancements in RAG technologies, their application in specialized sporting goods remains underdeveloped. Current major sports retailers (e.g., Pro:Direct Soccer, Sports Direct) utilize advanced filtering systems (facets) for product discovery, but they lack interactive, consultative interfaces.

Strengths of Current Solutions: Faceted search is highly efficient for users who already know exactly what they want (e.g., filtering by "Nike", "Size 10", "Firm Ground").

Weaknesses and Limitations: These systems fail when the user lacks technical domain knowledge. For instance, a parent buying boots for a child may not know the difference between 'Artificial Grass' (AG) and 'Turf' (TF) outsoles. Current platforms rely on static buying guides or external blogs, forcing the user to leave the shopping funnel to conduct research (Bawack et al. 2022).

Methodological Flaws in Existing Bots: Where chatbots are deployed on sports retail sites, they are almost exclusively used for customer service (post-purchase) rather than pre-purchase technical consultation.

4.4 Justification and Identified Gap

The literature reveals a distinct gap: while the MERN stack is proven for scalable e-commerce, and RAG AI is proven for intelligent information retrieval, there is a lack of integrated solutions designed specifically for consultative selling in highly technical, niche retail environments like football equipment. General-purpose e-commerce platforms do not natively map their product

schemas to vector embeddings for real-time conversational assistance. This project directly addresses this gap by engineering a bespoke MERN application where the MongoDB product catalogue is intrinsically linked to an AI embedding pipeline, effectively acting as a virtual technical specialist for amateur athletes.

5. Proposed Project Development and Methodology

5.1 Methodology Selection

Several software development methodologies exist for web engineering, most notably traditional sequential models (like Waterfall) and iterative models (like Agile and Spiral). According to Sommerville (2015), the Waterfall methodology requires rigid, comprehensive upfront requirement definitions, making it highly inflexible to changes later in the development cycle. Conversely, Agile methodologies prioritize iterative delivery, continuous testing, and adaptability.

For this project, an Agile (Scrum-based) methodology will be utilized. This approach is the most appropriate because integrating Artificial Intelligence (specifically tuning vector embeddings and conversational prompts) is inherently experimental and requires continuous refinement based on testing outputs (Beck et al. 2001). Breaking the project into two-week "sprints" allows for the core e-commerce platform to be built and tested first, functioning as a stable foundation before the complex AI microservices are integrated.

5.2 Tools and Technologies

The project will be built using the MERN stack, supplemented by specific AI APIs. Every tool has been selected based on industry standards for scalable web applications.

Component	Choice	Justification
Frontend Framework	React.js (with Vite)	Industry standard for Single Page Applications (SPAs); offers highly reusable UI components and fast hot-module replacement (HMR) during development.
State Management	Redux Toolkit	Provides predictable global state management, which is critical for complex e-commerce features like asynchronous shopping cart updates and user session handling.
Backend Environment	Node.js & Express.js	The non-blocking, event-driven architecture is highly efficient for handling simultaneous RESTful API calls and external AI API requests (Kaur and Kaur, 2022).
Database	MongoDB	A NoSQL document database allows for highly flexible, schema-less data structures, easily accommodating variable football equipment attributes (e.g., sizes, stud types).
AI Integration	Google Gemini API	Provides robust, pre-trained Large Language Models (LLMs) and tools to create a Retrieval-Augmented Generation (RAG) pipeline for the chatbot.

Payment Gateway	PayPal Developer API	Offers a secure, well-documented, and sandboxed environment for simulating e-commerce transactions safely.
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5.3 Data Management

The system will manage three primary categories of data:

Product Data: A custom dataset of football equipment (boots, gloves, accessories) will be compiled and stored in MongoDB. Images will be hosted via a cloud service (Cloudinary) to reduce server load.

User Data: Customer information, order histories, and authentication details will be securely stored. Passwords will be heavily encrypted using bcrypt.js prior to database insertion to ensure security compliance.

Vector Embeddings: To facilitate the AI chatbot, product descriptions will be converted into mathematical vector representations (embeddings) and stored. When a user asks a question, the system will query these embeddings to fetch the most relevant product context for the Gemini AI model to read before responding.

5.4 Development Plan

Following the Agile methodology, the development lifecycle will be divided into the following sequential phases:

Phases	Phase's names	Tasks
Phase 1	Design & Architecture	UI wireframing, database schema design, and API endpoint planning.
Phase 2	Core Backend	Development of RESTful APIs, authentication middleware, and database connections.
Phase 3	Frontend E-commerce Integration	Frontend E-commerce Integration: Building the React interface, Redux store setup, cart logic, and PayPal integration.
Phase 4	AI & Vector Database Implementation	Scripting the embedding generation and building the conversational chatbot route using the Gemini API.
Phase 5	Testing & Evaluation	End-to-end integration testing, user acceptance testing (UAT), and final bug squashing.

6. Project Scope and Feasibility

6.1 Project Scope

To prevent scope creep and ensure the successful delivery of a functional Minimum Viable Product (MVP) within the academic timeframe, the boundaries of this project have been strictly defined.

In-Scope (Will Deliver)	Out-of-Scope (Won't Cover)
Responsive web application (Desktop & Mobile Browser)	Native mobile application (iOS/Android apps)
User authentication and secure session management	Social media single sign-on (SSO) integrations
Core e-commerce: Product catalogue, cart, and checkout	Complex warehouse/inventory logistics management
Simulated secure payment processing via PayPal Sandbox	Live, real-money financial transactions
Admin dashboard for CRUD operations (Products/Orders/Users)	Automated marketing or email campaign systems
AI Chatbot utilizing Gemini API for product recommendations based on store inventory	Live human customer support chat integration

6.2 Feasibility Analysis

The project's viability has been evaluated across the three core dimensions of the feasibility triangle: Time, Technical capability, and Resources.

Time Feasibility: The project is scheduled for completion over a standard 12-week academic timeline. The Agile methodology selected in Section 5 inherently supports this by prioritizing the delivery of core e-commerce features in early sprints. To mitigate unforeseen delays (such as API integration issues), a strict 15% buffer time has been incorporated into the project plan, ensuring the timeline remains realistic and achievable.

Technical Feasibility: The technical complexity of this project is high but manageable. I currently possess foundational skills in JavaScript, React.js, and Node.js. While integrating the Gemini API and generating vector embeddings represents a new learning curve, Google provides extensive, well-documented SDKs for Node.js. Furthermore, restricting the AI strictly to the store's text data (via RAG) simplifies the machine learning requirement, making it an achievable challenge rather than an insurmountable barrier.

Resource Feasibility: The project is highly feasible from a resource perspective as it relies almost entirely on open-source or free-tier cloud services.

Software/Hardware: Development will occur on a standard personal computer using VS Code (free).

Data: The product dataset (images, descriptions, pricing for football gear) will be manually compiled from publicly available retailer data, eliminating the need to purchase proprietary datasets.

Hosting & APIs: MongoDB Atlas provides a free cloud database tier. The Gemini API offers generous free-tier access suitable for development and testing, and PayPal provides free developer sandbox accounts. Therefore, the project requires zero financial capital to execute.

7. Project Evaluation and Success Criteria

7.1 Evaluation Approach

To ensure the project's aim is met, the system will be evaluated across two distinct technical domains: the core web application (e-commerce functionality) and the integrated artificial intelligence (Gemini-powered chatbot).

The web application's usability will be evaluated through empirical user testing with at least 10 participants, utilizing the industry-standard System Usability Scale (SUS) to quantify the user experience (Brooke 1996). The technical performance of both the MERN backend and the Gemini API will be evaluated through rigorous automated testing and latency benchmarking.

7.2 Measurable Success Criteria

The following quantifiable metrics have been established to define the successful delivery of the project. Vague assessments (e.g., "the application works well") have been intentionally avoided in favor of strict, testable targets.

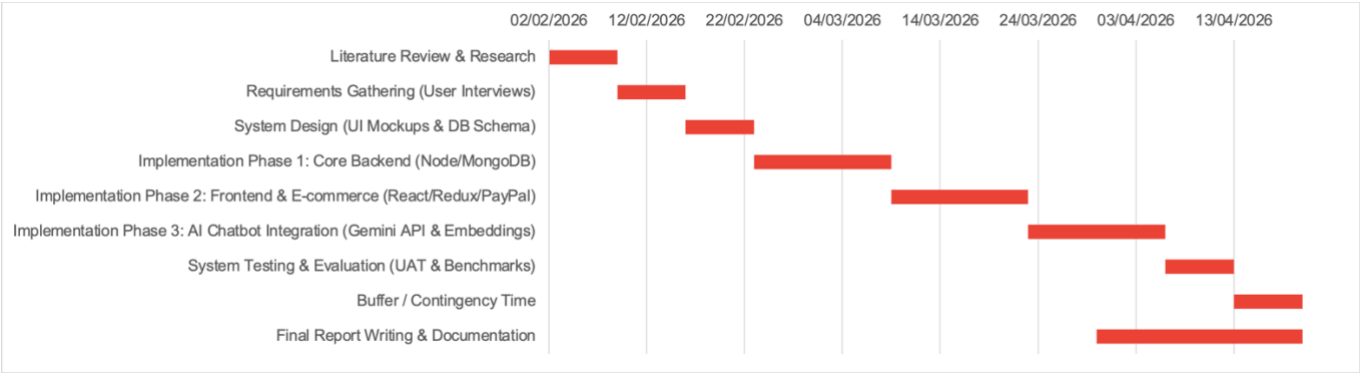
Criterion	Evaluation Metric	Target / Success Threshold
User Satisfaction & Usability	System Usability Scale (SUS) Score	Average score > 75 across 10 test users.
Core Functionality	Task Completion Rate	100% success rate for critical user flows (e.g., User Registration, Add to Cart, Sandbox Checkout).
System Performance (Web)	Average Page Load & API Latency	Backend REST API response times < 500ms; First Contentful Paint (FCP) < 2.0 seconds.
AI Chatbot Accuracy	Contextual Relevance Score	> 85% of AI responses accurately rely ONLY on the store's provided product data (no hallucinated inventory).
AI Chatbot Latency	API Response Time (Gemini + RAG)	AI text generation and retrieval latency < 3.0 seconds per user query.
Code Quality & Reliability	Unit/Integration Test Coverage	> 80% test coverage for core backend models (Products, Users, Orders) using testing frameworks like Jest.

8. Project Plan and Timeline

8.1 Project Approach and Dependencies

The project will be executed over a standard 12-week academic timeframe using an Agile (Scrum-based) methodology. To manage dependencies, the project is structured sequentially: the database schema and API endpoints (Phase 1) must be functional before the React frontend (Phase 2) can be built. Similarly, the core e-commerce catalogue must be populated before the Gemini AI vector embeddings (Phase 3) can be generated. To mitigate unforeseen technical delays, a contingency buffer of approximately 15% (nearly 2 weeks) has been built into the final stages of the timeline, running concurrently with report writing and final evaluations. Regular bi-weekly meetings with the project supervisor are scheduled to ensure the project remains aligned with academic standards.

8.2 Proposed Gantt Chart Schedule



9. Expected Outcomes and Contributions

9.1 Expected Deliverables

Upon the successful completion of this project, the following tangible artifacts will be delivered:

Software Application: A fully functional, responsive, MERN-stack e-commerce web application, featuring a complete user purchasing journey (from product browsing to simulated secure checkout) and an administrative dashboard for inventory management.

AI Microservice: An integrated conversational AI assistant powered by the Google Gemini API, utilizing a Retrieval-Augmented Generation (RAG) architecture to answer domain-specific questions based exclusively on the store's football equipment database.

Technical Documentation: Comprehensive system documentation, including the MongoDB schema design, API endpoint specifications, and an installation manual.

Academic Research Report: A final written dissertation detailing the project's methodology, critical analysis of the development lifecycle, and the empirical results of the user testing (SUS scores and latency benchmarks).

9.2 Project Contributions

The primary contribution of this project lies in its practical application of Artificial Intelligence within a specialized retail context. As identified in the Literature Review (Section 4), there is a distinct gap in the current e-commerce landscape: while generic platforms handle transactions efficiently, they fail to provide the highly technical, consultative guidance required for specialized sporting goods (such as interpreting boot stud configurations or glove latex types).

By intrinsically linking a NoSQL product database to an AI embedding pipeline, this project addresses that gap. The resulting application serves as a proof-of-concept demonstrating how niche online retailers can cost-effectively deploy Large Language Models (LLMs) to mimic expert human customer service. Consequently, this work advances web engineering practices by showcasing a modern, scalable approach to building highly personalized, context-aware digital storefronts that prioritize user education alongside transactional efficiency.

10. LSEPI Considerations and Risks

10.1 LSEPI Considerations

Category	Key Considerations & Project Application
Legal	Data Protection: The system will collect user data (names, emails, addresses, and order histories). This necessitates compliance with the General Data Protection Regulation (GDPR). All passwords will be cryptographically hashed using bcrypt before database storage. Payment Security: To comply with financial regulations, no live credit card data will be stored on the server; all transactions will be handled via the secure PayPal API.
Social	Accessibility: The web interface must be designed following Web Content Accessibility Guidelines (WCAG) to ensure visually impaired users can navigate the product catalogue using screen readers. Digital Divide: The AI chatbot assumes a level of digital literacy and high-speed internet access, which may exclude some older or rural demographics from utilizing the platform fully.
Ethical	AI Bias and Hallucination: A significant ethical concern with Large Language Models (LLMs) is algorithmic bias or hallucinated information. The Gemini API must be strictly constrained via system prompting to only recommend products based on factual database inputs, preventing it from making unfounded or unsafe claims about sports equipment preventing injuries.
Professional	Code of Conduct: The project will be conducted in accordance with the British Computer Society (BCS) Code of Conduct, ensuring honesty in the evaluation phase. Test results (e.g., SUS scores and latency benchmarks) will be reported objectively, without manipulating data to falsely achieve success criteria.

10.2 Project Risks and Mitigation Strategies

Risk Category	Specific Risk Description	Impact / Likelihood	Mitigation Strategy
Technical (Critical)	The Gemini API integration suffers from high latency, or the RAG pipeline fails to accurately retrieve product embeddings.	High Impact Medium Likelihood	Start AI development early in Phase 4. Implement fallback UI mechanisms (e.g., standard search) so the core e-commerce site functions even if the AI fails.
Time (High)	Implementation of the complex Redux state management for the	Medium Impact High Likelihood	Utilize Agile methodology to prioritize MVP features first. A 2-week buffer period has

	shopping cart takes longer than anticipated.		been explicitly scheduled to absorb these delays.
Data (Medium)	Insufficient or low-quality product descriptions prevent the AI from generating useful text embeddings.	High Impact Low Likelihood	Manually curate and verify a localized dataset of 50+ high-quality product descriptions prior to writing the embedding script.
Resource (Low)	Exceeding the free-tier data limits on MongoDB Atlas or Cloudinary during testing.	Low Impact Medium Likelihood	Compress all product images before upload and periodically flush dummy user data from the database during the testing phase.

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